



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD 2613
SYSTEM ANALYSIS AND DESIGN

PROJECT : PHASE 3
Analysis and Design

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1.0 Overview of the Project

In this project, we will collect information requirements and look at system requirements for this project. This project will determine the information needs that can help you grasp the 4W1H of an operational system. To gather the information, we interviewed the client with some questions about workflow and condition of the current system. After gathering the requirements information, we have a solid understanding of the business function based on a piece of precise information about the people, objectives, data, and procedures involved.

The business flow of the AS-IS system for task management is then visualized using context diagrams in the following steps which are described as a high-level Data Flow Diagram (DFD), a context diagram depicting the information flow between the taskmanagement.com and external entities, including users and lecturers.

Data flow diagrams, which are the parent and child diagrams, are used to depict the external entities, data flows, data stores, and procedures involved once the context diagram has been created. Parent diagram is known as a level 0 diagram which is the explosion of context diagram. An integer is assigned to each process. The parent diagram includes all external entities as well as the main data storage. Each process in the parent diagram will be exploded to generate a child diagram. A child diagram utilizes the same number as the parent diagram, receives the same input, and produces the same output.

When the tasks are finally completed, we now thoroughly understand the current system and data flow diagrams, entity relationship diagrams, and information requirements.

2.0 Problem Statement

2.1 Fragmented Task Management

Postgraduates struggle with fragmented task management processes, such as manually organizing research materials, coordinating with co-authors, and tracking deadlines for paper submission.

2.2 Communication Gaps

Communication breakdowns between students and supervisors, as well as among co-authors, result in misunderstandings, missed deadlines, and delays in the publication process.

2.3 Resource Allocation Issues

Difficulty in allocating resources effectively, such as time, research materials, and collaboration tools, leads to inefficiencies and suboptimal outcomes in research and publication endeavors.

2.4 Deadline Management

Poor management of deadlines for literature review, data collection, analysis, and writing stages hampers progress and increases the likelihood of rushed, incomplete submissions.

2.5 Collaboration Challenges

Coordinating tasks and sharing progress among co-authors or research team members becomes cumbersome due to reliance on disparate communication channels and tools.

2.6 Publication Requirement Compliance

Lack of awareness or reminders regarding publication requirements, formatting guidelines, and submission procedures results in unnecessary revisions and rejections from journals or conferences

3.0 Proposed Solutions

Task Management System (TMS) is a system that allows postgraduate students to manage their assignments and projects efficiently. The system can be installed and applied on mobile devices such as mobile phones and tablets in order to enhance the functions of the Task Management System (TMS).

The Task Management System's (TMS) first function assists users or postgraduate students in allocating assignments so they can schedule a time to finish the projects. Apart from that, users can simply adhere to the timelines while working on projects because the system automatically saves the work timelines into a spreadsheet file each time a user opens and closes a project file. Additionally, the users are able to monitor the status of each work and the projects' progress. The users are required to enter the task progression from 0% to 100%. As a result, users can determine if the task is finished or only halfway done.

The second feature is the specialized collaboration and communication tools. The co-author of a project can be added by the author or postgraduate students who collaborate on the project. While the partners are now offline, users can still discuss projects with them by leaving a message in the communication space. As soon as they get the notification, the partners can reply to the message.

The third feature is the notification feature. The system will notify the users when they received messages, the upcoming due date, the existence of overdue tasks. The system helps users manage their time so they can complete their project publication and keeps the projects on schedule.

Besides that, the Task Management System uses a mobile platform, it requires a mobile gadget and Internet access to use the system. A server and database system are required for this system since the system has to store user projects' progression and communicate with the team members. A security system guards the system to prevent confidential project information from being discovered by unidentified third parties.

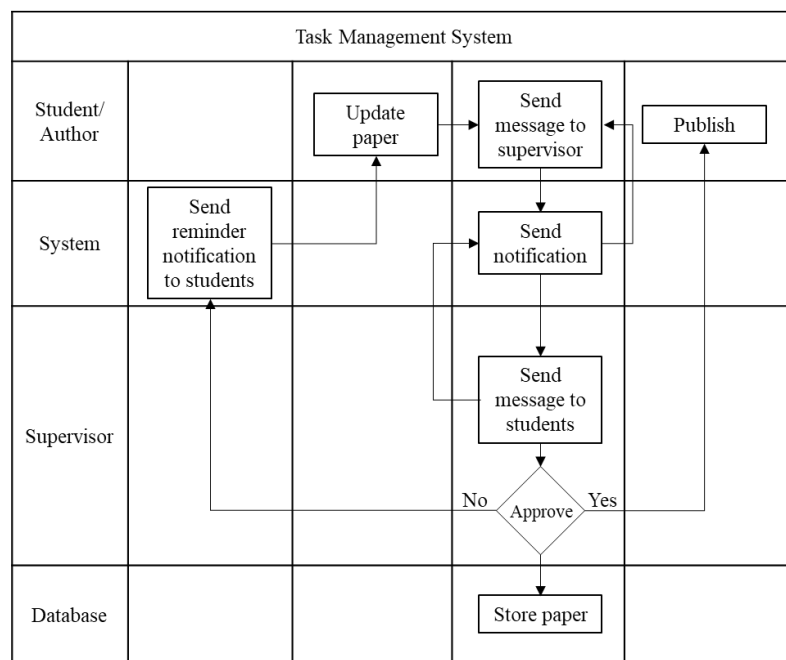
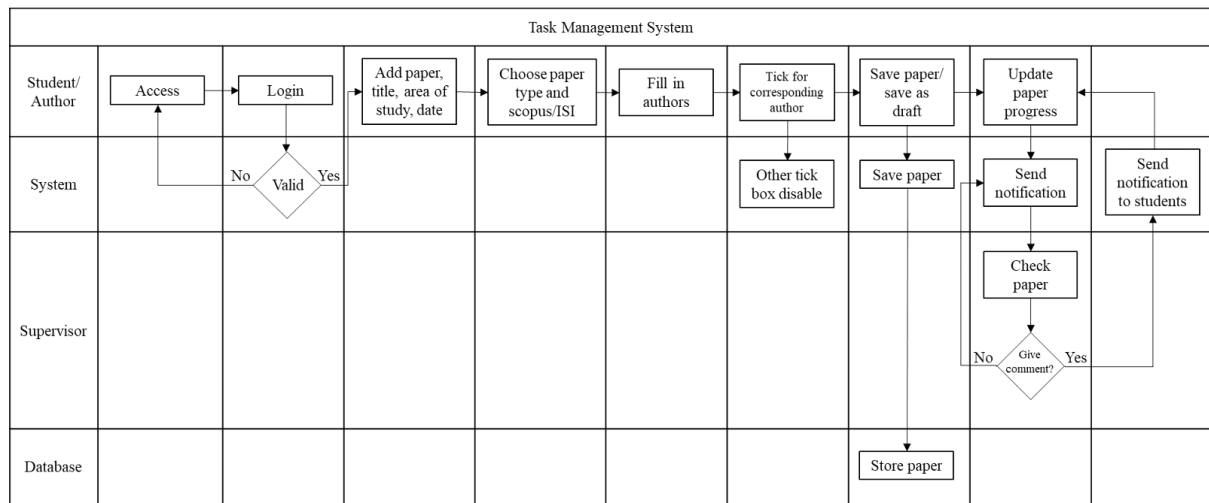
Furthermore, the Task Management System requires permission to access dates, times, and files to track the progress of the project and manage the file. The message and project timeline are stored in a database within the system. Users can search the message history for messages sent by other team members. All the information and data are password-protected, the data can be accessed by the users with the password only.

4.0 Current business process (scenarios, workflow)

Below are the scenarios and workflow on how the task management system current business process:

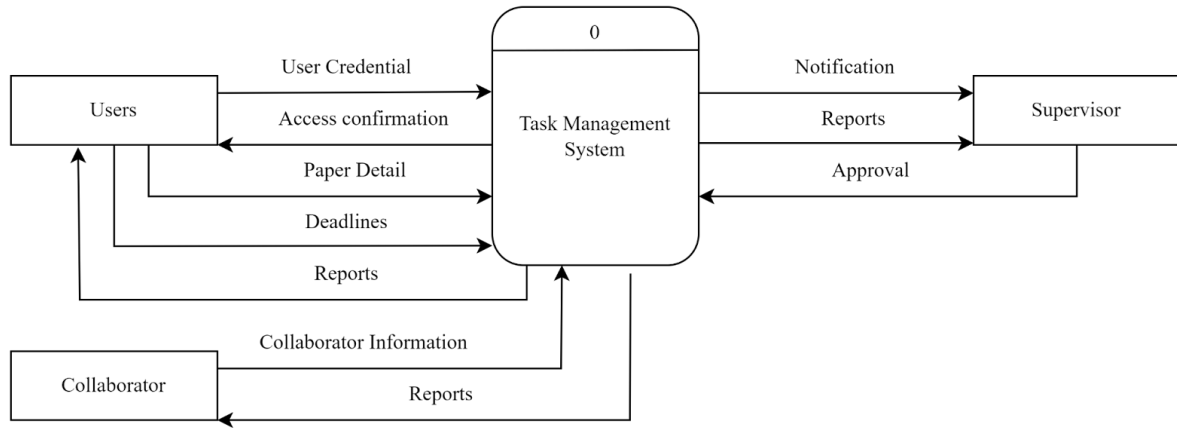
1. Users access their account using their university email.
2. Users log in to the system.
 - 2.1 If users fail to login, users are required to attempt login again.
3. Students can add paper and fill in their title, area of study and choose the date from the calendar.
4. Choose the paper type and scopus/ISI.
5. Fill in the first author and co-author up to 13 people (if required).
6. Students need to tick for the corresponding author in the tick box.
 - 6.1 Once there is a tick box that has been ticked, the other tick box will be disabled.
7. Then students can choose to save the paper or save it as a draft.
8. After students have added paper, they can update their paper progress so that they won't miss out any phases.
 - 8.1 Once the students update the progress, there will be a notification sent to their supervisor and also other authors so that everyone can always follow up the progress/changes of the paper.
 - 8.2 There will be a calendar in the dashboard that records the progress and deadlines of the paper for all dates and times so that students can refer easily.
9. In this system, there will also be a reminder feature. When the deadline is near, the system will automatically send notification to the students to notify them to update their paper on time.
 - 9.1 Supervisor will also be notified to check student's papers if the supervisor did not give any comment 3 days after the student's update.
10. Students can always message their supervisor for any question in this system so that their messages will not be obscured by other messages.
 - 10.1 Students and supervisors will receive the notification of the messages immediately so that they won't miss out on any messages.
11. Once the students have done the paper and get approval from the supervisor, they can prepare to publish their paper.

Workflow:

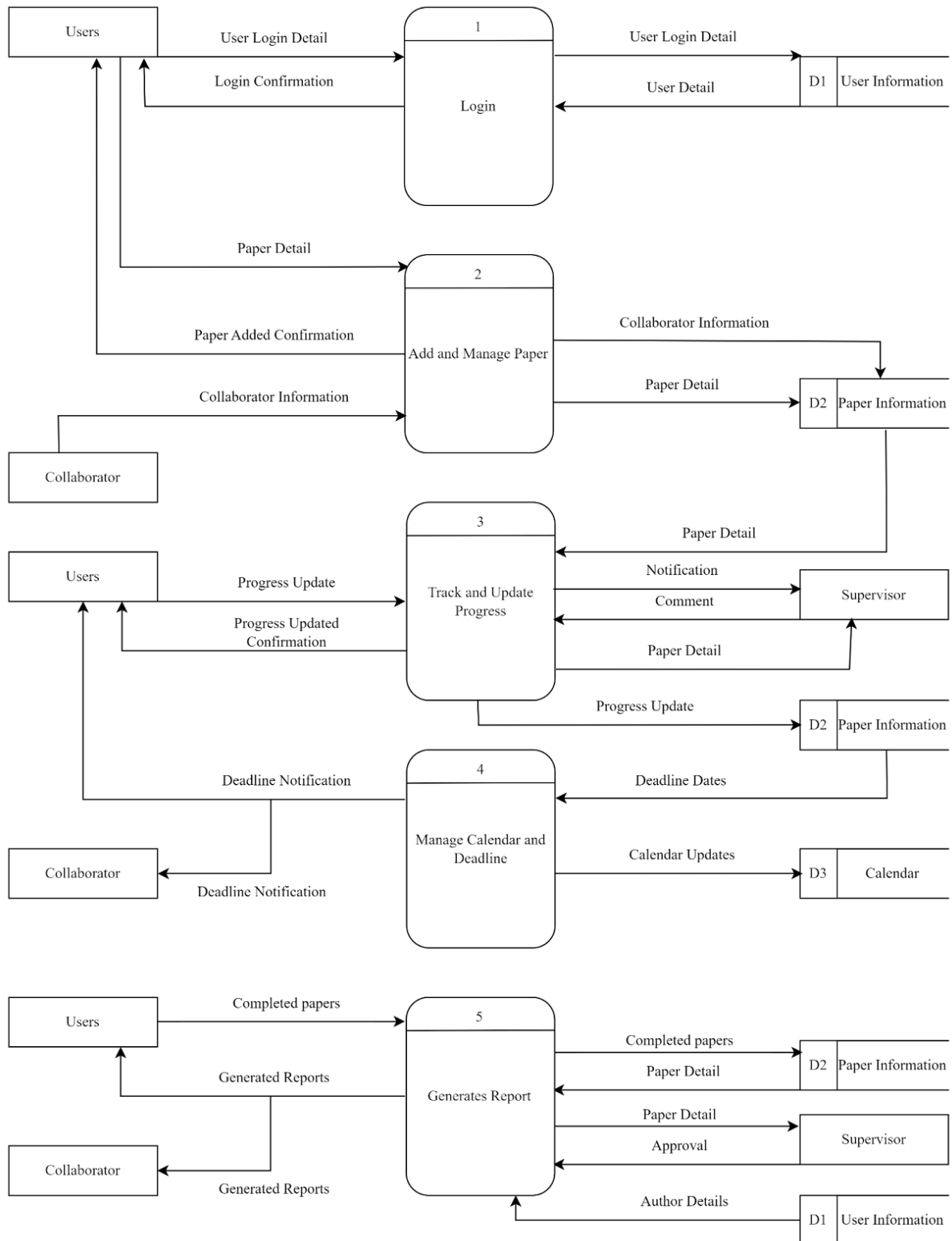


5.0 Logical DFD AS-IS system

5.1 Context Diagram

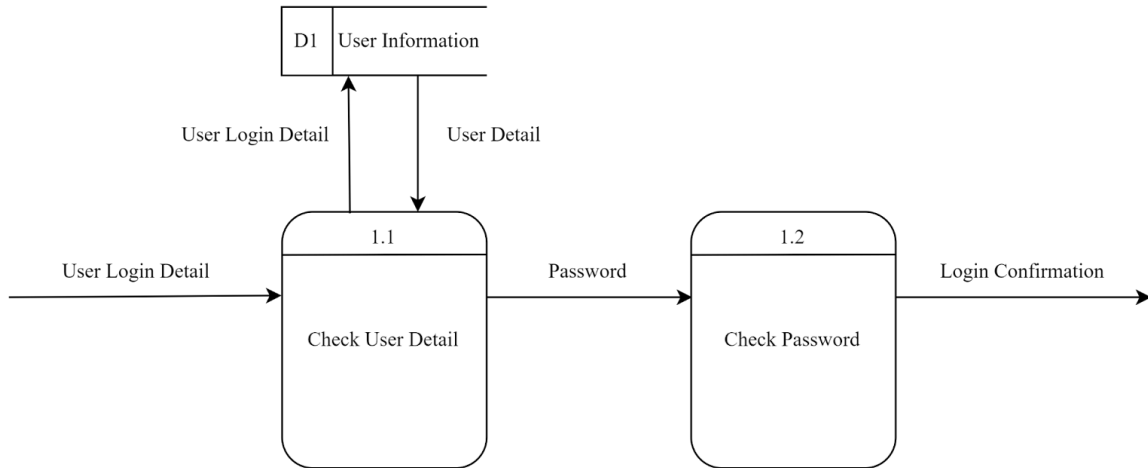


5.2 Level 0 Diagram

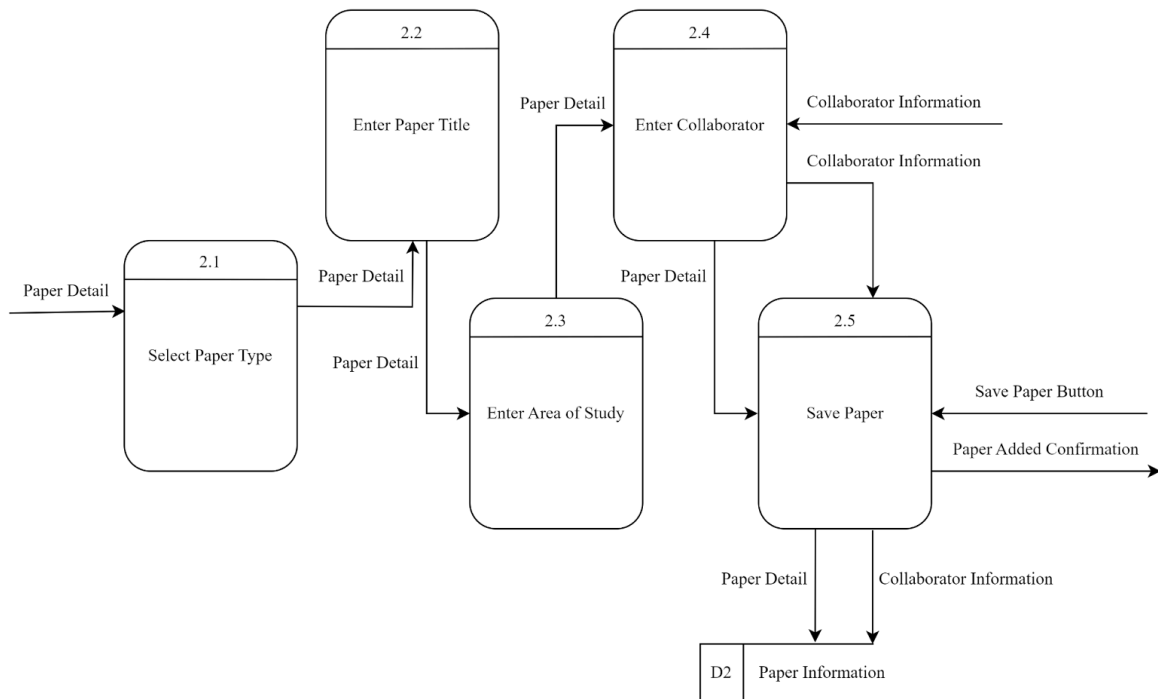


5.3 Child Diagram

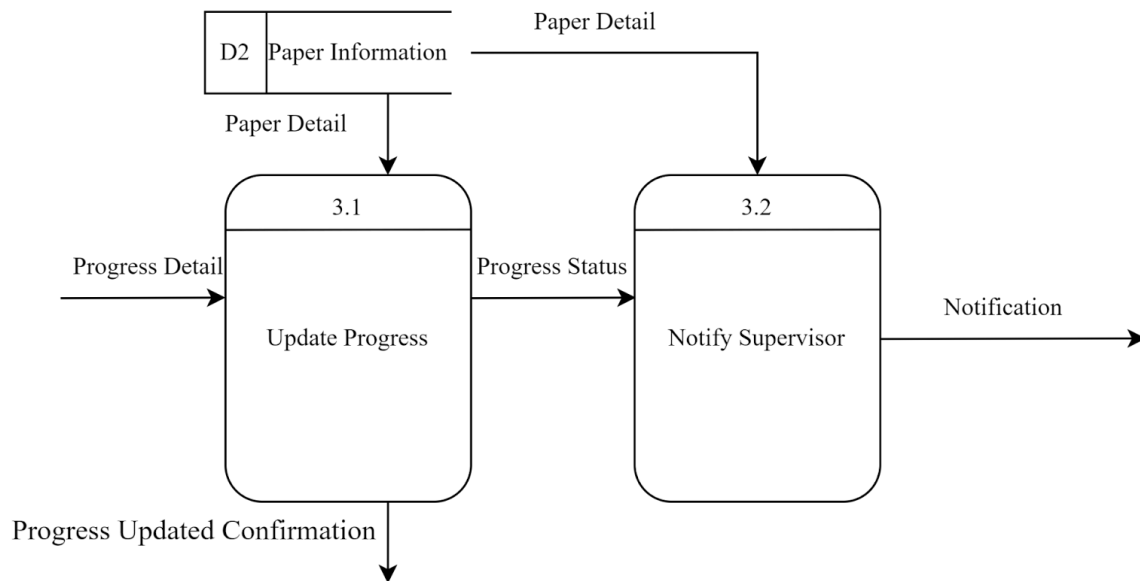
5.3.1 Login



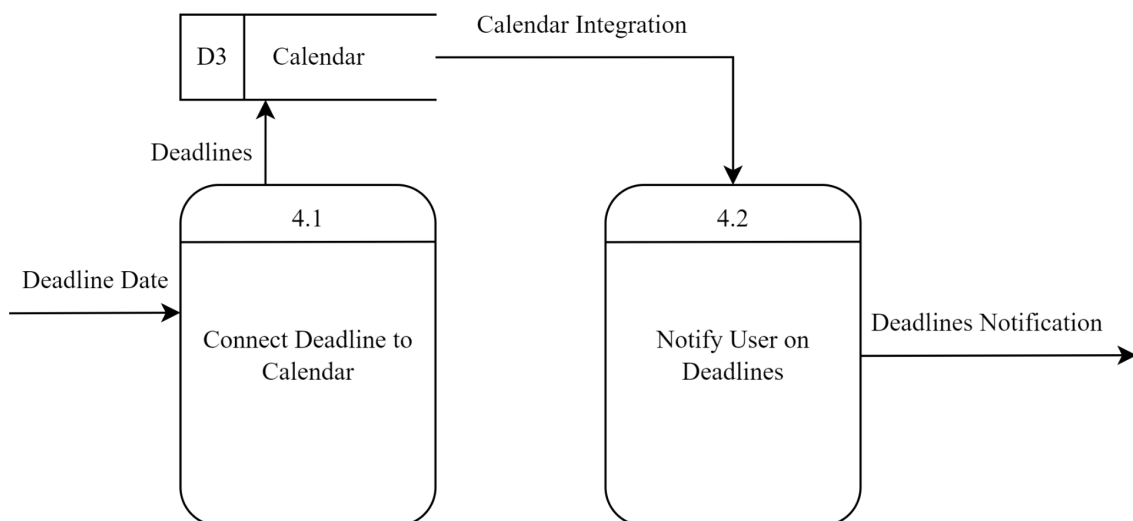
5.3.2 Add and Manage Paper



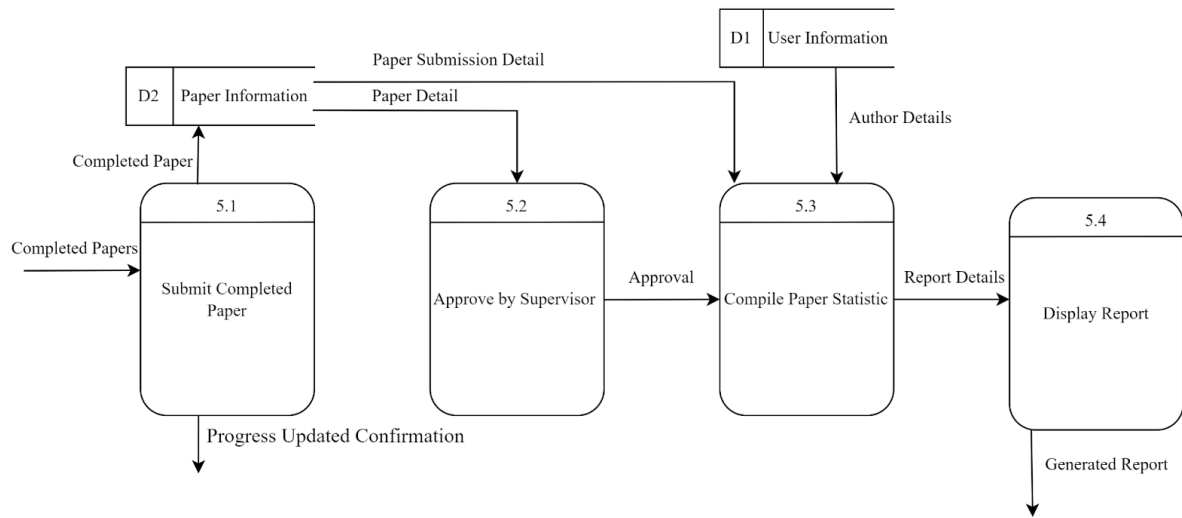
5.3.3 Track and Update Progress



5.3.4 Manage Calendar and Deadlines



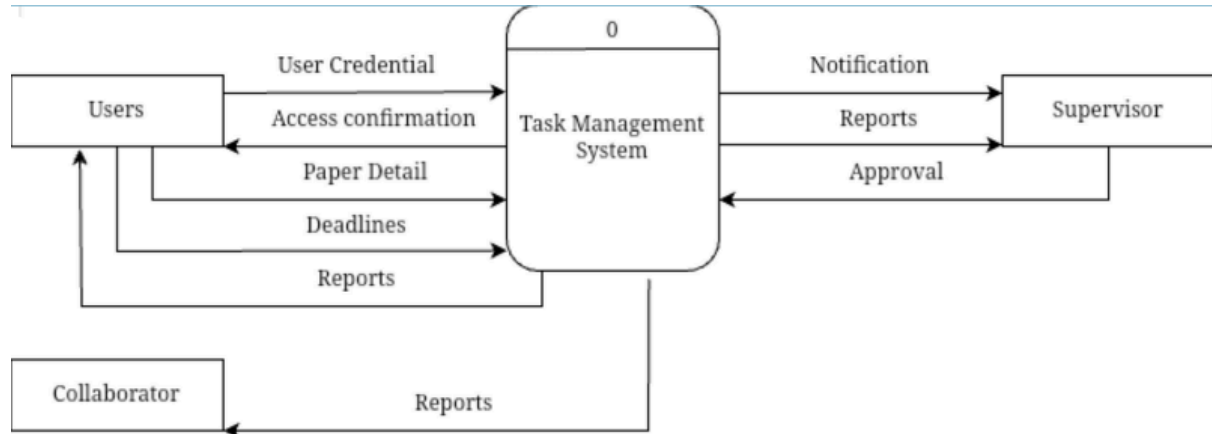
5.3.5 Generate Reports



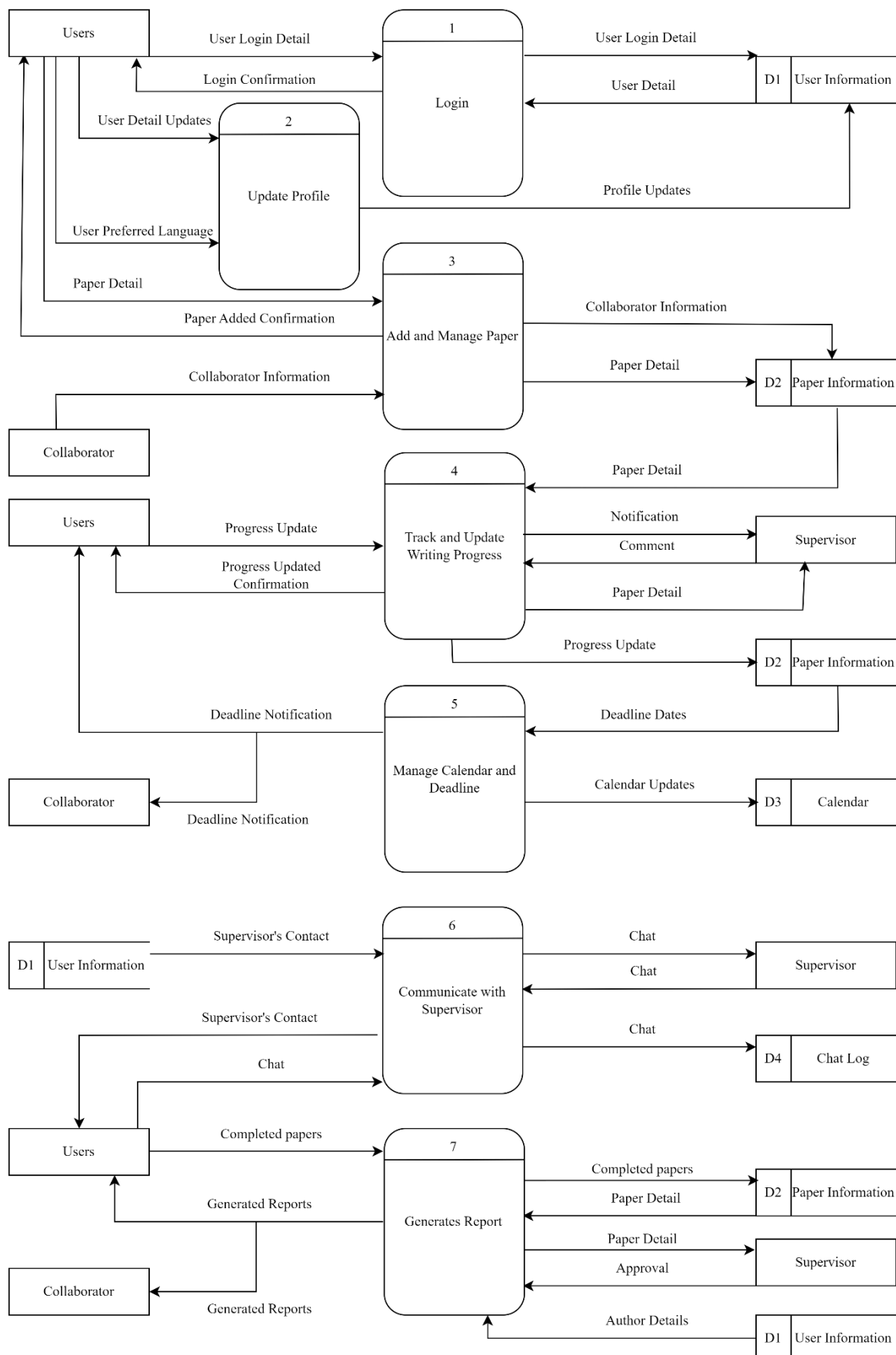
6.0 System Analysis and Specification

6.1 Logical DFD TO-BE system

6.1.1 Context Diagram

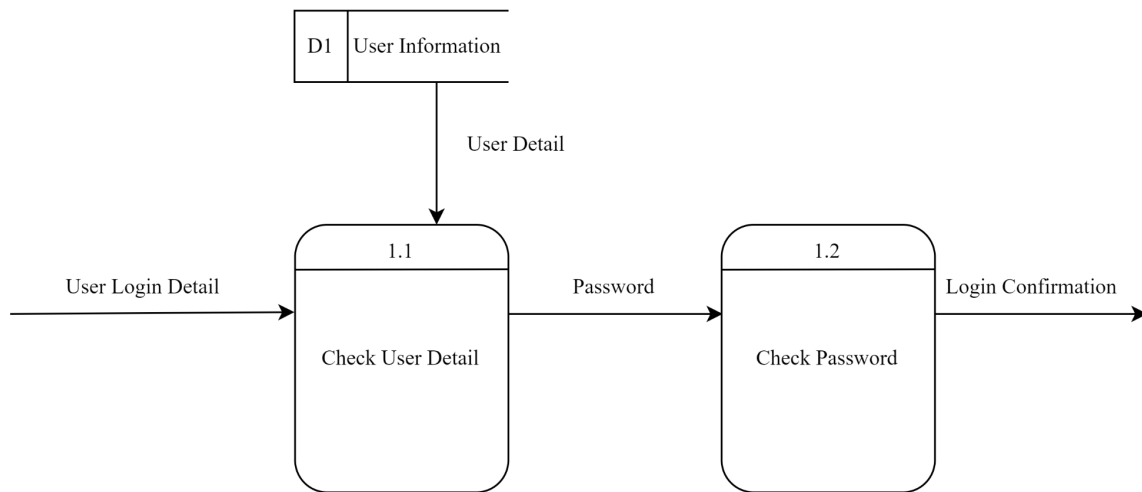


6.1.2 Level 0 Diagram

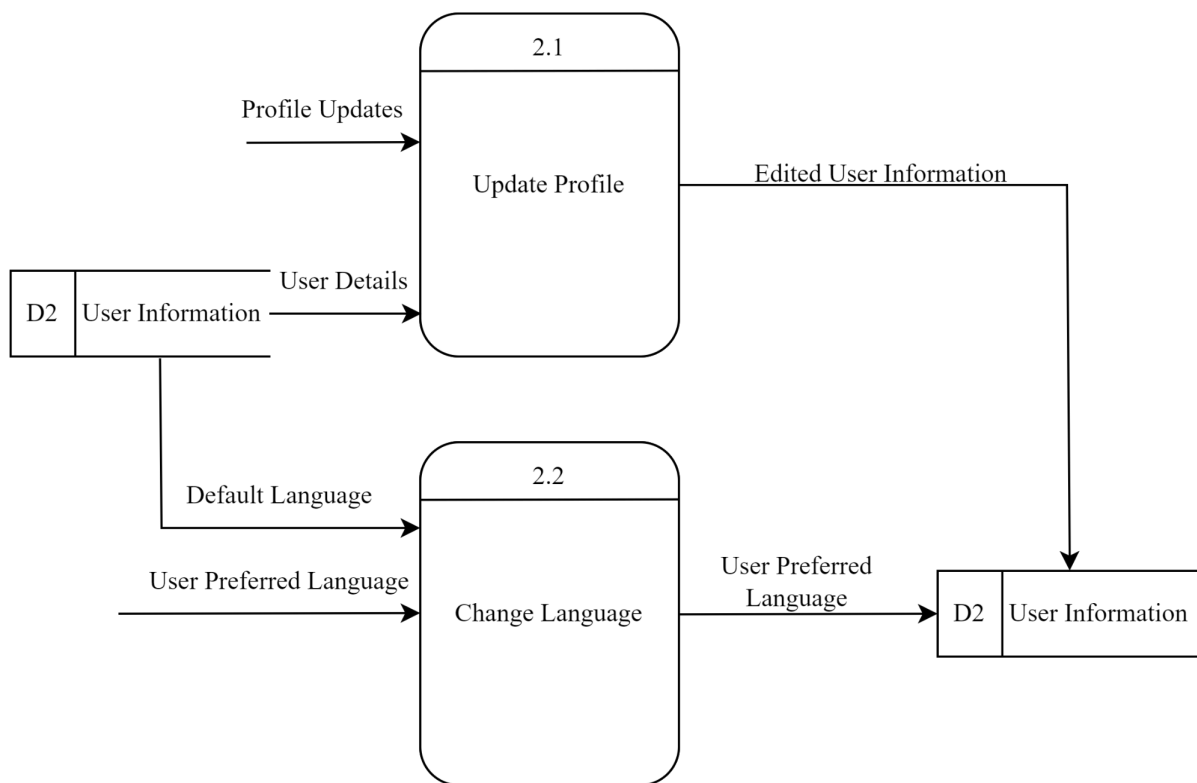


6.1.3 Child Diagram

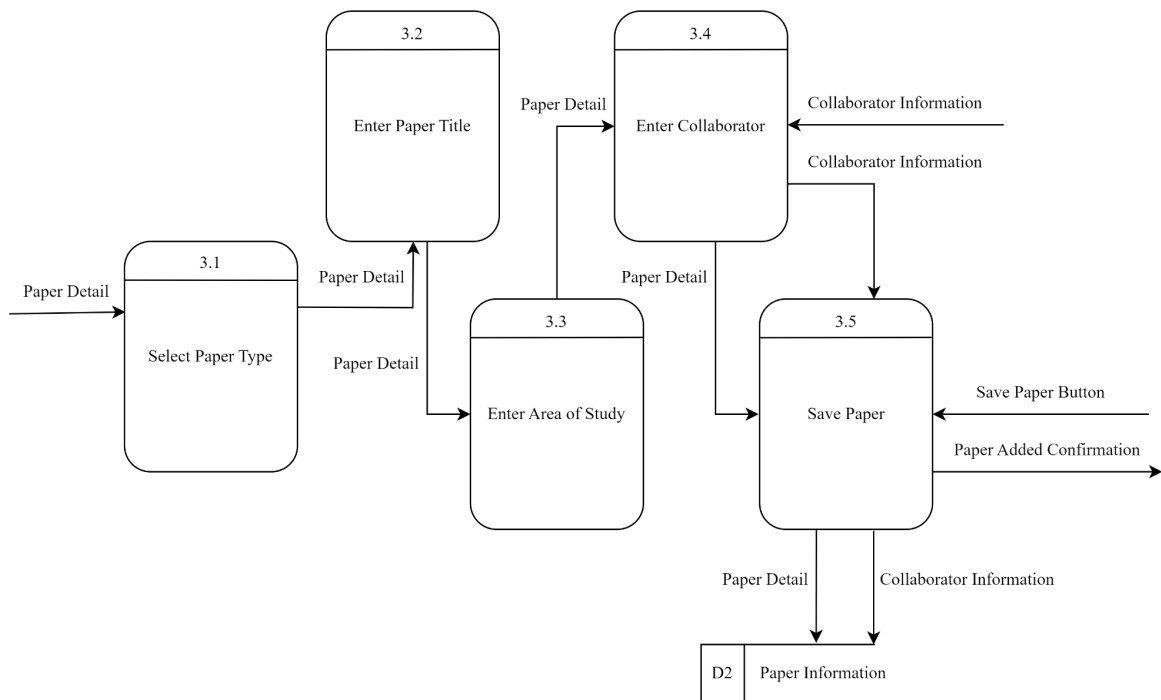
6.1.3.1 User Login



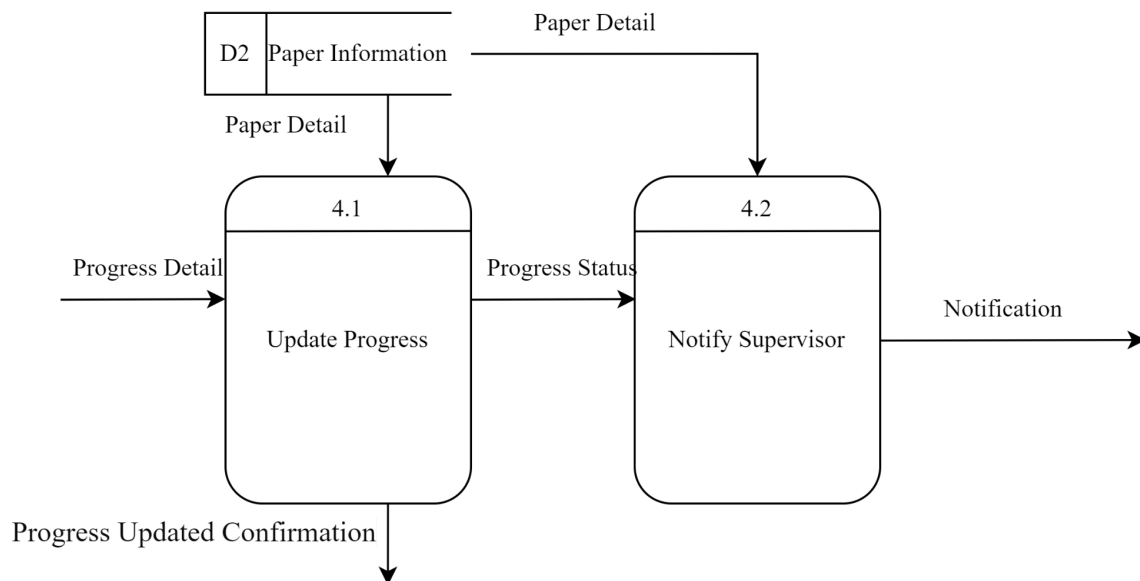
6.1.3.2 User Profile



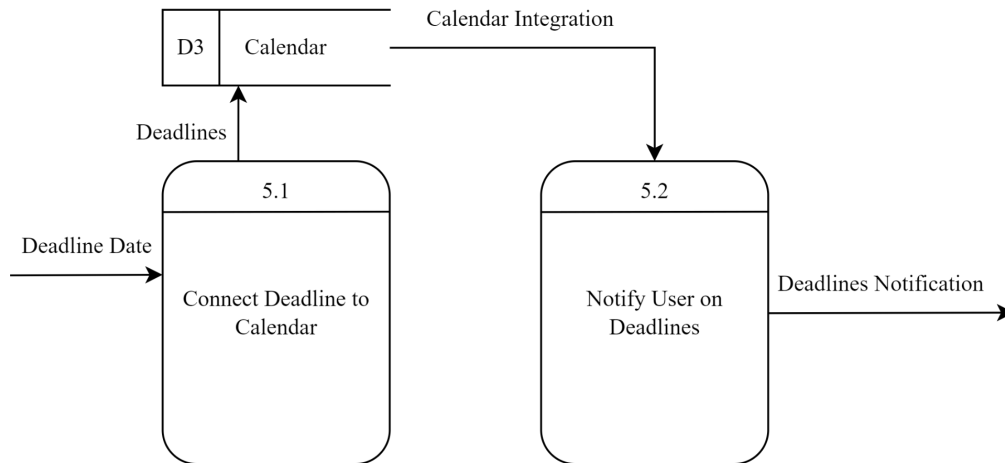
6.1.3.3 Add and Manage Paper



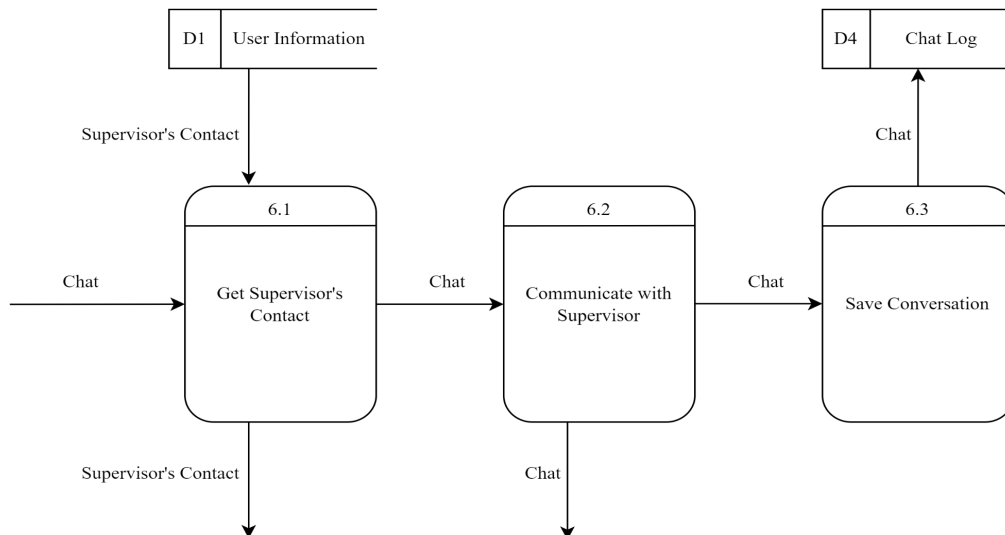
6.1.3.4 Track and Update Writing Progress



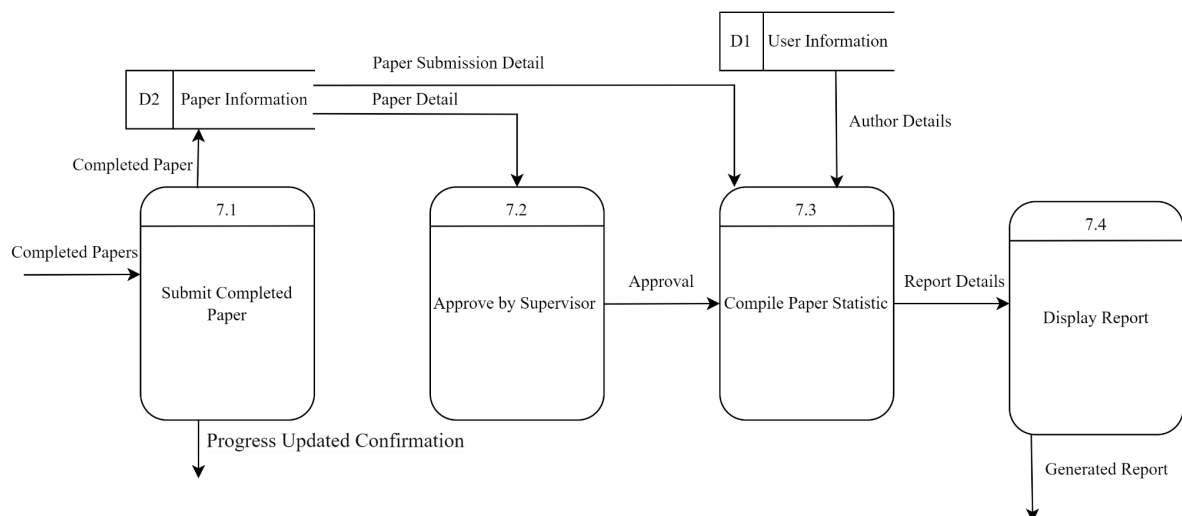
6.1.3.5 Manage Calendar and Deadlines



6.1.3.6 Communicate with supervisor



6.1.3.7 Generate Report



6.2 Process Specification (based on Logical DFD TO-BE)

Structured English is used to model and illustrate the process of the Logical TO-BE system based on the Logical DFD TO-BE.

6.2.1 User Login

```
DO
GET user login details
GET user details from database
CHECK user details and password
BEGIN IF
IF received valid login detail
    THEN proceed to next step
ELSE IF received invalid login detail
    THEN inform user the login account is invalid
END IF
END
```

6.2.2 Update Profile

```
DO
GET user details from database
PERFORM user details
BEGIN IF
IF user details incorrect
    THEN edit user information
    STORE user information into database
END IF
BEGIN IF
IF user not familiar with the language
    THEN change language in settings
    STORE system settings into database
END IF
END
```

6.2.3 Add and Manage Paper

```
DO
GET paper detail
STORE paper detail into database
BEGIN IF
IF want to manage paper writing
    THEN manage paper
    GET paper detail from database
    EDIT paper detail
END IF
END
```

6.2.4 Track and Update Writing Progress

```
DO
User update writing progress
STORE progress into database
BEGIN IF
IF want to do writing progress tracking
    THEN track progress
    GET progress from database
    READ progress
END IF
END
```

6.2.5 Manage Calendar and Deadlines

```
DO
GET deadline from database
STORE deadline into calander
GET deadlines notification
END
```

6.2.6 Communicate with Supervisor

```
DO
GET supervisor's contact from database
COMMUNICATE with supervisor
STORE conservation into database
```

6.2.7 Generate Report

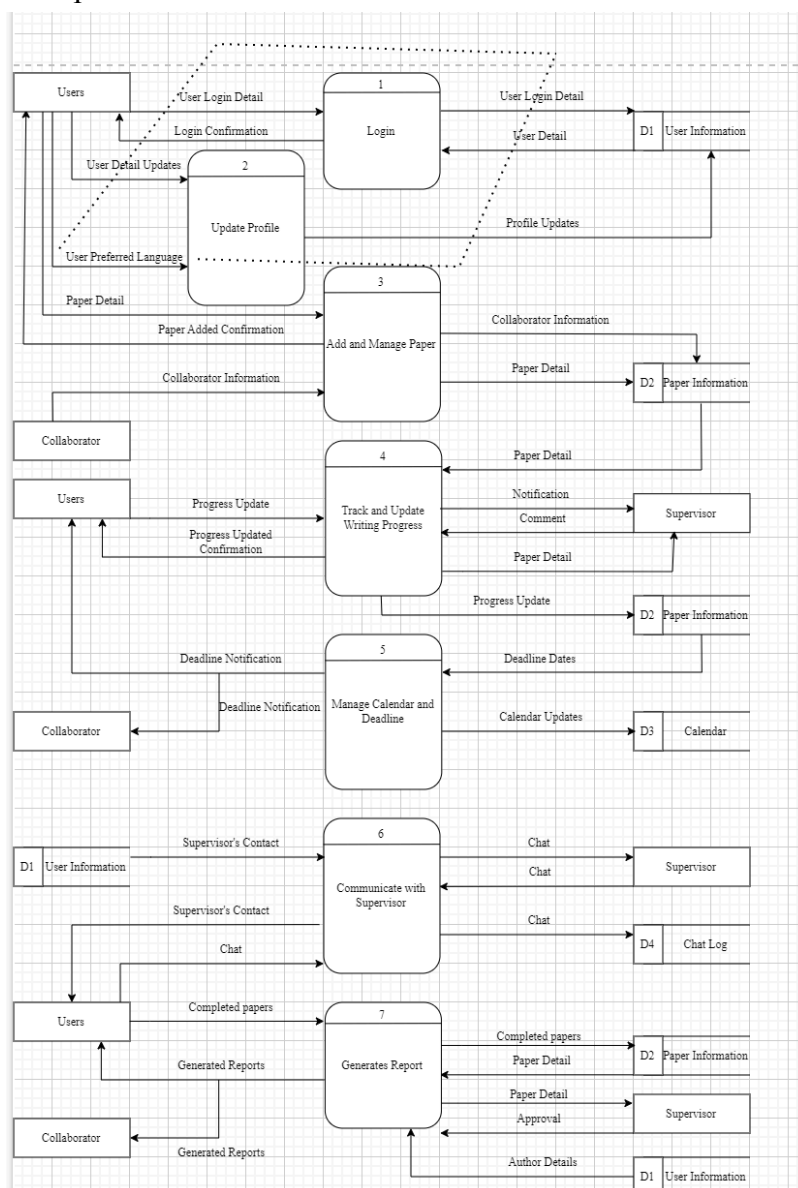
```
DO
SUBMIT completed paper
STORE paper details into database
BEGIN IF
IF approved by supervisor
    THEN compile paper statistics
    DISPLAY generated report
ELSE IF rejected by supervisor
    THEN inform user rewrite the paper
END IF
END IF
END
```

7.0 Physical System Design

7.1 Physical DFD TO-BE system (Diagram 0, Child, Partitioning, CRUD Matrix, Event Response Table, Structure Chart, System Architecture)

7.1.1 Physical Level 0 Diagram

In this partitioned physical DFD, the first partitioning is the first process because it is manually executed by users. Users need to log in to the system by school official id/email and password. The system appears to be designed for managing user profiles, paper submissions, and interactions between users, collaborators, and supervisors. Here's a breakdown of the main processes:



Login and Update Profile: Processes 1 and 2 handle user authentication and profile management. Users can log in, update their profiles, and set preferred languages. User information is stored in the D1 User Information data store.

Paper Management: Process 3 allows users and collaborators to add and manage papers. Paper details are stored in the D2 Paper Information data store.

Progress Tracking: Process 4 enables users to track and update writing progress. Supervisors can review paper details and provide comments. Progress updates are stored in the D2 Paper Information data store.

Calendar and Deadline Management: Process 5 manages calendars and deadlines. It notifies users and collaborators about deadlines and updates the D3 Calendar data store.

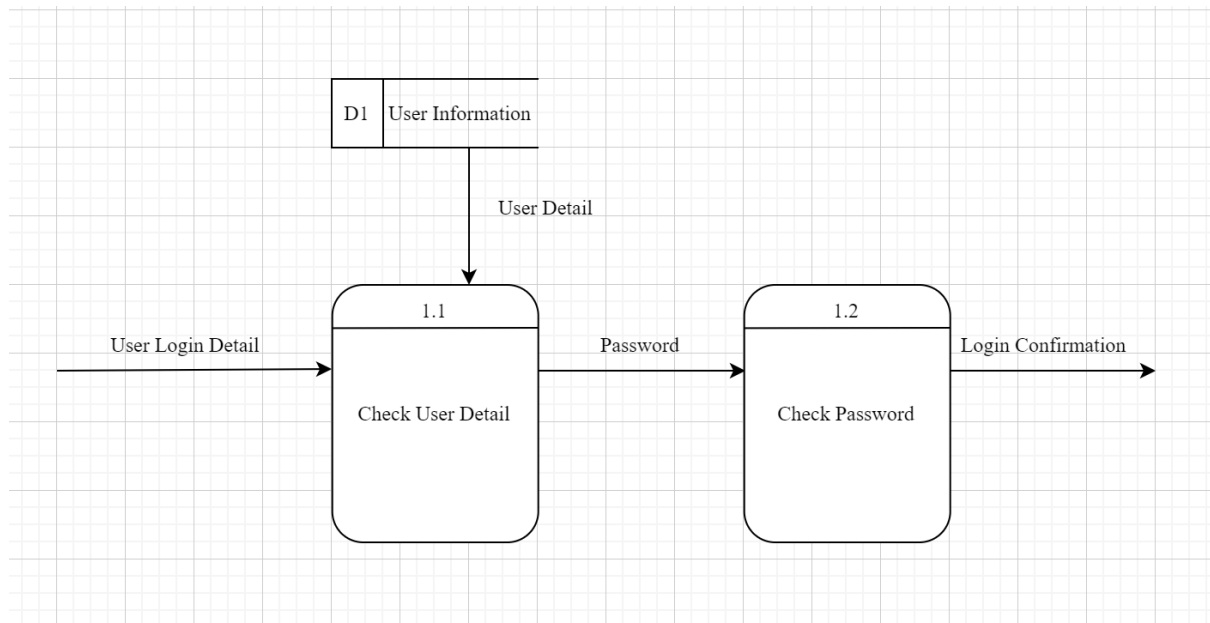
Communication: Process 6 facilitates communication between users and supervisors. Chat logs are stored in the D4 Chat Log data store.

Report Generation: Process 7 generates reports based on completed papers. It interacts with users, collaborators, and supervisors, updating the D2 Paper Information data store.

The system integrates multiple user roles (Users, Collaborators, Supervisors) and data stores to manage the workflow of paper writing, submission, and review processes. It provides features for user management, paper tracking, progress monitoring, deadline management, and communication between different stakeholders.

7.1.2 Physical Child Diagram

7.1.2.1 Login



External Entity:

D1 User Information: This represents the data store containing user information.

Processes:

1.1 Check User Detail: This process verifies the user's login details.

1.2 Check Password: This process validates the user's password.

Data Flows:

User Detail: Flows from D1 to process 1.1, providing user information.

User Login Detail: Input to process 1.1, likely containing username or other identifier.

Password: Flows from process 1.1 to 1.2, passing the password for verification.

Login Confirmation: Output from process 1.2, indicating whether login was successful.

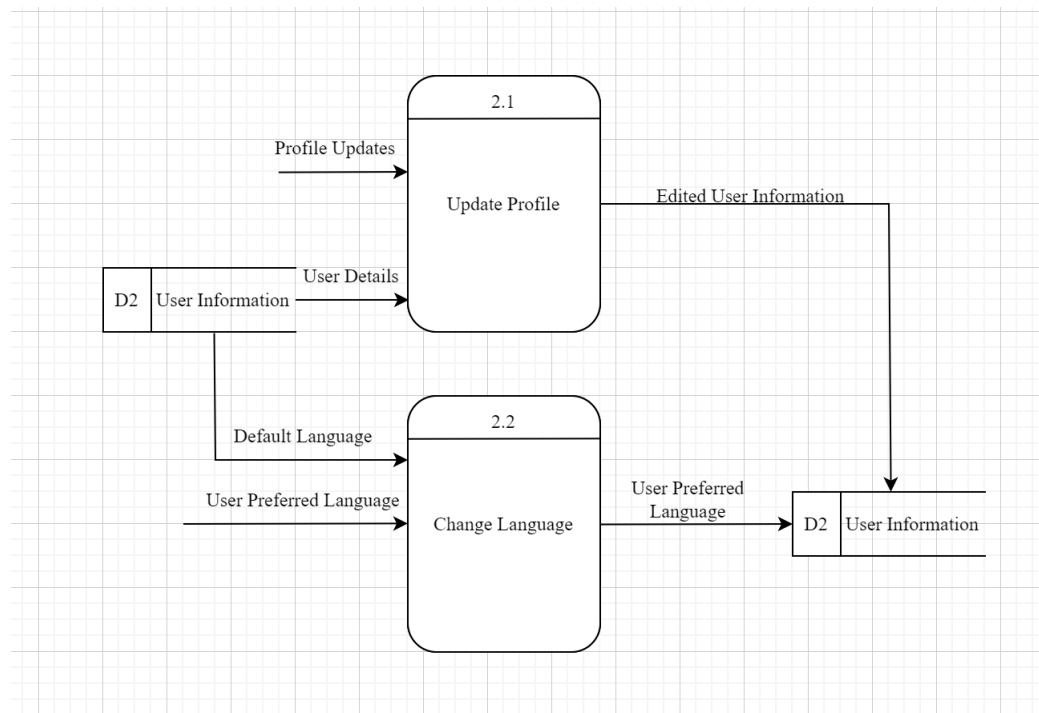
User login details are input to the "Check User Detail" process.

This process retrieves user information from the data store.

If the user is found, the password is passed to the "Check Password" process.

The "Check Password" process validates the password and produces a login confirmation.

7.1.2.2 Manage Profile



Data Store:

D2 User Information: This represents the database storing user information.

Processes:

2.1 Update Profile: Handles profile updates.

2.2 Change Language: Manages language preference changes.

Data Flows:

Profile Updates: Input to process 2.1 for updating user profiles.

User Details: Flows from D2 to process 2.1, providing current user information.

Edited User Information: Output from process 2.1 back to D2, storing updated profile data.

Default Language: Flows from D2 to process 2.2, providing the system's default language.

User Preferred Language: Input to process 2.2 for changing language settings.

User Preferred Language (output): Flows from process 2.2 to D2, updating the user's language preference.

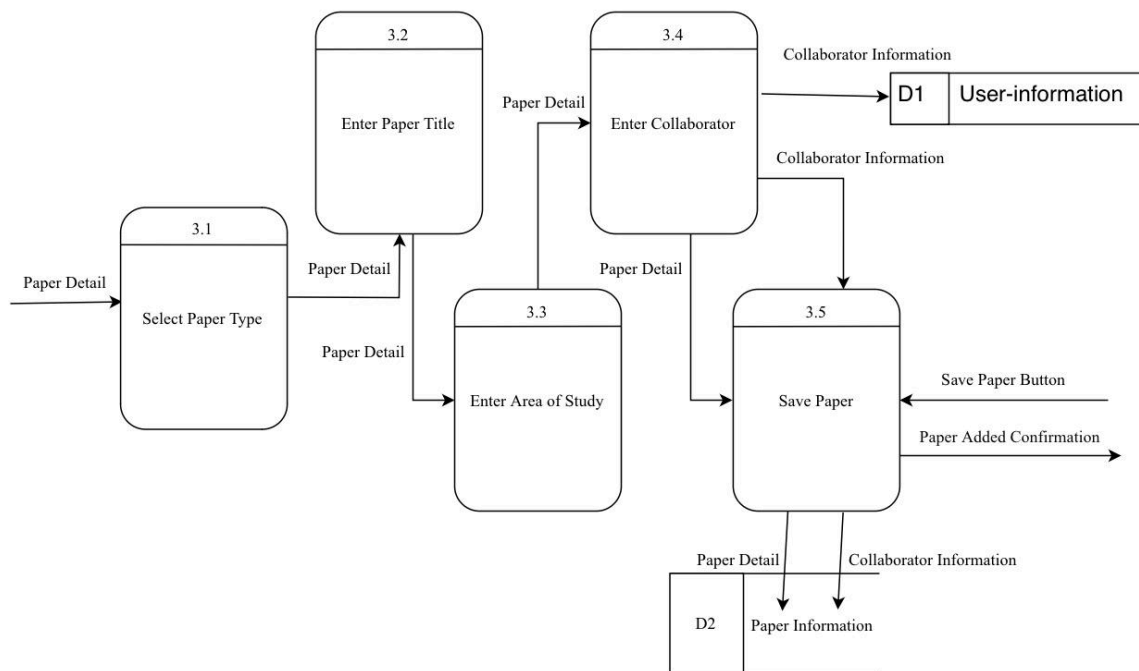
This DFD illustrates two main user profile management functions:

Profile updating, where users can modify their information.

Language preference setting, allowing users to change their preferred language.

Both processes interact with the User Information data store, reading from and writing to it as needed. This diagram provides a clear view of how user data is managed and updated in the system.

7.1.2.3 Add and Manage Paper



The process starts at 3.1, "Select Paper Type". This is where the user begins by choosing the type of paper.

From there, "Paper Detail" flows to three different processes:

3.2: "Enter Paper Title"

3.3: "Enter Area of Study"

3.4: "Enter Collaborator"

The "Enter Collaborator" process (3.4) also receives "Collaborator Information" as input.

All these details then flow into process 3.5, "Save Paper". This process also receives input from a "Save Paper Button".

The "Save Paper" process has three outputs:

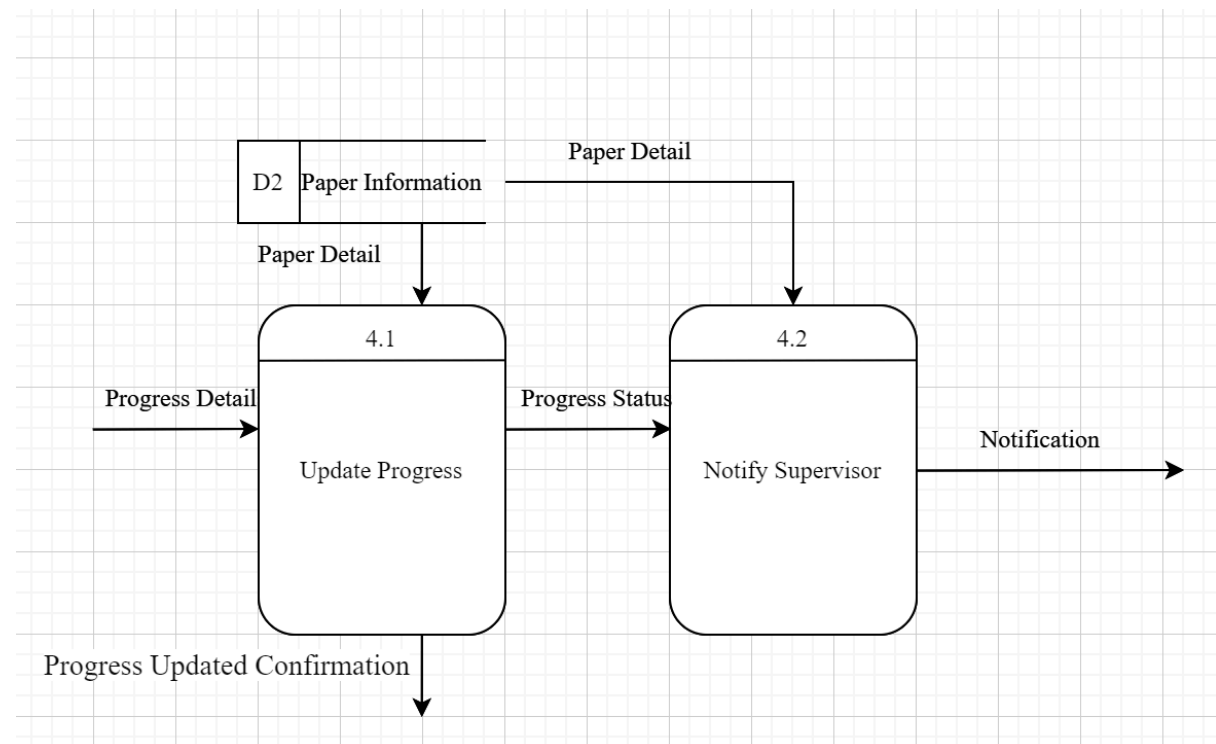
"Paper Added Confirmation", likely a message to the user

"Paper Detail" and "Collaborator Information", which flow to "D2 Paper Information", presumably a data store where the paper's information is saved

"Paper Detail", which flows back to the "Select Paper Type" process, possibly for adding another paper or for reference

This diagram represents a systematic approach to capturing and storing information about academic papers, including their type, title, area of study, and collaborators. The numbered processes (3.1 to 3.5) suggest this might be part of a larger system with other components not shown in this particular diagram.

7.1.2.4 Track and Upload Progress



This data flow diagram illustrates a process related to updating and notifying about paper progress. This is a data store containing information about papers.

Process 4.1 - Update Progress:

Receives "Paper Detail" from D2 Paper Information

Takes "Progress Detail" as an input

Outputs "Progress Updated Confirmation"

Sends "Progress Status" to Process 4.2

Process 4.2 - Notify Supervisor:

Receives "Paper Detail" from D2 Paper Information

Receives "Progress Status" from Process 4.1

Outputs "Notification"

The flow of information is as follows:

Paper details flow from the D2 Paper Information store to both processes 4.1 and 4.2.

Progress Detail is input into Process 4.1 to update the progress.

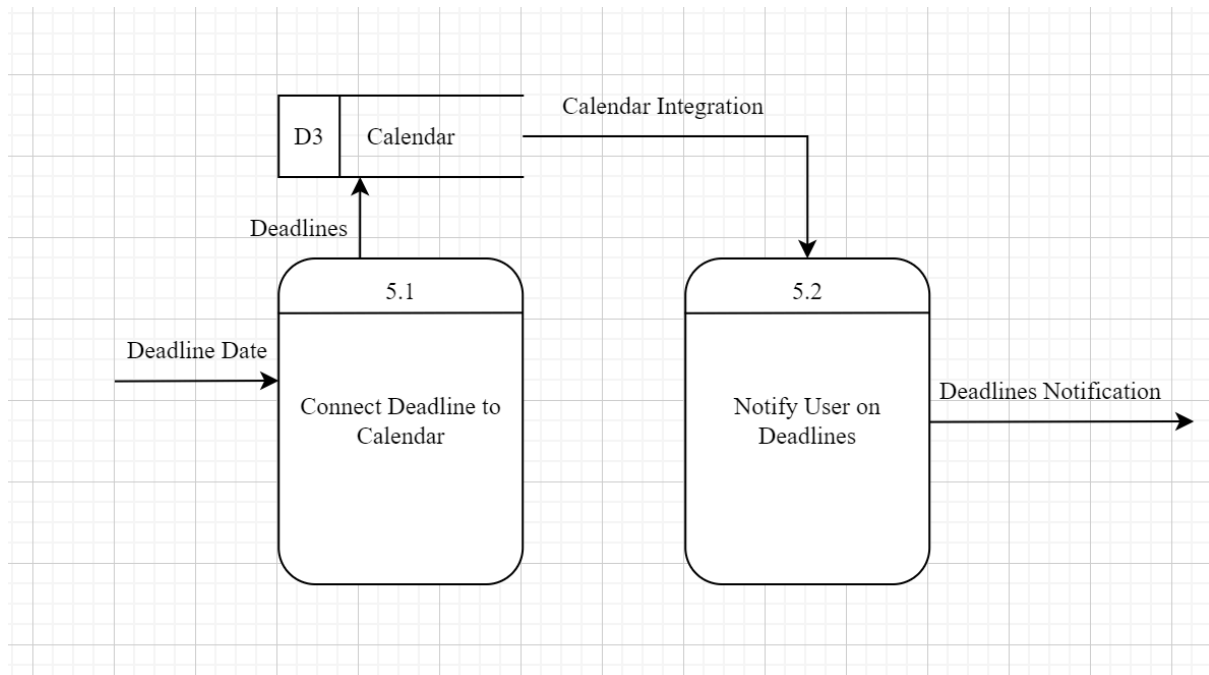
Process 4.1 outputs a Progress Updated Confirmation.

Process 4.1 sends the Progress Status to Process 4.2.

Process 4.2 combines the Paper Detail and Progress Status to Notify the Supervisor.

This diagram represents a system where paper progress is updated, and then a supervisor is notified of the updated status, likely in an academic or publishing context.

7.1.2.5 Manage Calendar and Deadlines



D3 Calendar: This is a data store containing calendar information.

Process 5.1 - Connect Deadline to Calendar:

Receives "Deadline Date" as an input

Sends "Deadlines" to the D3 Calendar data store

Process 5.2 - Notify User on Deadlines:

Receives "Calendar Integration" information from D3 Calendar

Outputs "Deadlines Notification"

A "Deadline Date" is input into Process 5.1.

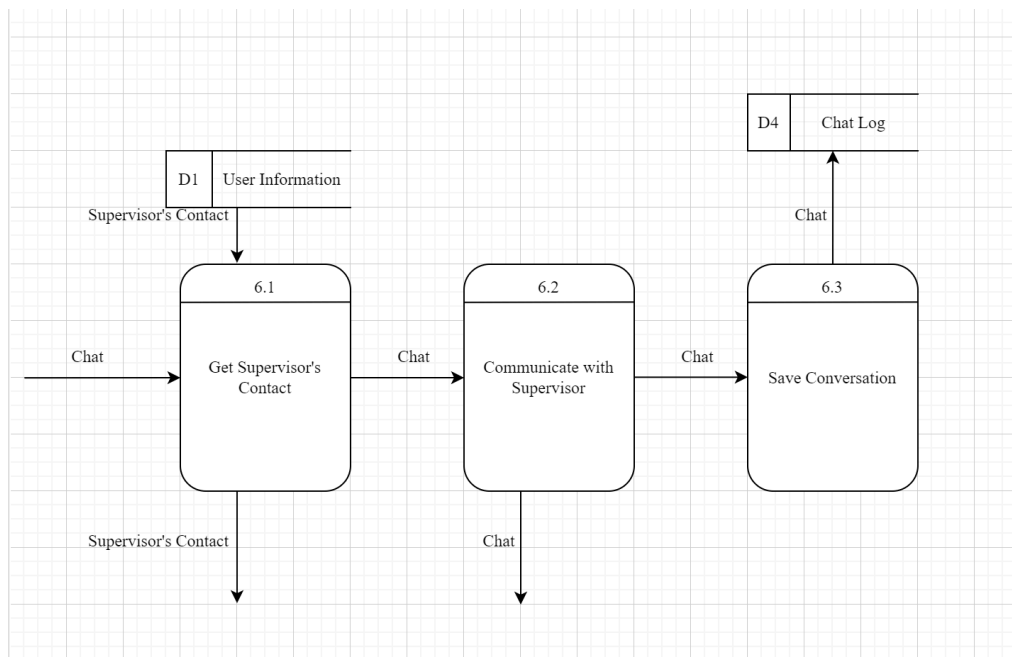
Process 5.1 connects this deadline to the calendar and sends the "Deadlines" to the D3 Calendar data store.

The D3 Calendar data store provides "Calendar Integration" information to Process 5.2.

Process 5.2 uses this information to notify the user about deadlines, outputting "Deadlines Notification".

This diagram represents a system where deadlines are added to a calendar, and then users are notified of these deadlines. This could be useful in various contexts such as project management, academic settings, or any environment where deadline tracking is important.

7.1.2.6 Communicate with supervisor



Data Stores:

D1: User Information

D4: Chat Log

Processes:

6.1: Get Supervisor's Contact

6.2: Communicate with Supervisor

6.3: Save Conversation

Flow:

- "Chat" input initiates Process 6.1.
- Process 6.1 retrieves "Supervisor's Contact" from D1 User Information.
- Process 6.1 outputs "Supervisor's Contact" and passes "Chat" to Process 6.2.
- Process 6.2 "Communicate with Supervisor" and outputs "Chat".
- The "Chat" from Process 6.2 is input to Process 6.3.
- Process 6.3 "Save Conversation" and stores the "Chat" in D4 Chat Log.

This system represents a communication workflow:

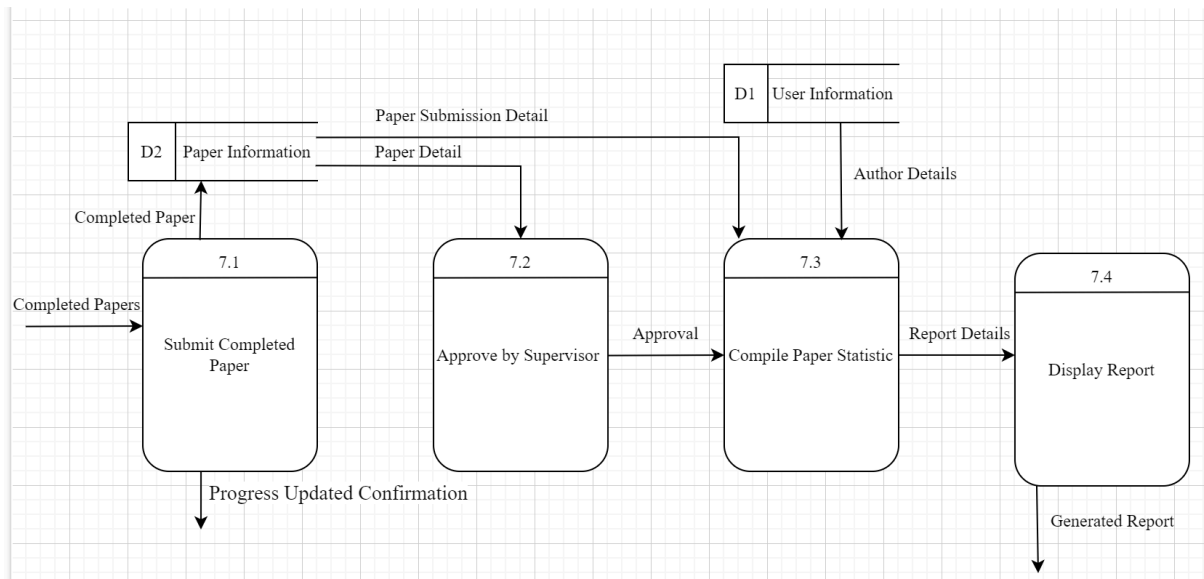
A user initiates a chat and the system retrieves the supervisor's contact information.

The communication takes place between the user and the supervisor.

The conversation is then saved in a chat log for record-keeping.

This workflow could be used in various organizational settings to facilitate and document communication between employees and their supervisors, ensuring that interactions are recorded and easily accessible for future reference.

7.1.2.7 Generate reports



Flow:

- "Completed Papers" are input into Process 7.1.
- Process 7.1 sends "Completed Paper" to D2 Paper Information.
- Process 7.1 outputs "Progress Updated Confirmation".
- D2 Paper Information sends "Paper Detail" to Process 7.2.
- Process 7.2 sends "Approval" to Process 7.3.
- D2 Paper Information also sends "Paper Submission Detail" to Process 7.3.
- D1 User Information sends "Author Details" to Process 7.3.
- Process 7.3 compiles statistics and sends "Report Details" to Process 7.4.
- Process 7.4 "Display Report" and outputs a "Generated Report".

This system represents an academic paper submission and reporting process:

Papers are submitted and stored.

Supervisors approve the papers.

Statistics are compiled using paper details, submission info, and author data.

A report is generated and displayed.

This workflow could be used in academic institutions or publishing houses to manage the paper submission, approval, and reporting process.

7.1.3 CRUD Matrix

Process	Data Store			
	User Information	Paper Information	Calendar	Chat Log
User Login	R			
Manage Profile	R, U			
Add and Manage Paper		C, R, U		
Track and Upload Progress		R, U		
Manage Calendar and Deadlines			C,R,U	
Communicate with supervisor	R			C
Generate reports	R	R		

The CRUD matrix for the proposed Task Management System shows the interaction between system processes and data stores. User Login processes read user information for authentication. User Profile processes both read and update user data. The Add and Manage Paper process handles creating, reading, and updating paper details. Tracking and updating writing progress involves reading and updating the paper information. Calendar and deadline management processes create, read, and update calendar entries. Communication with supervisors involves reading user information and creating chat logs. Finally, generating reports reads user and paper information to compile and present necessary data. This matrix helps in understanding the data flow and system functionality at a glance.

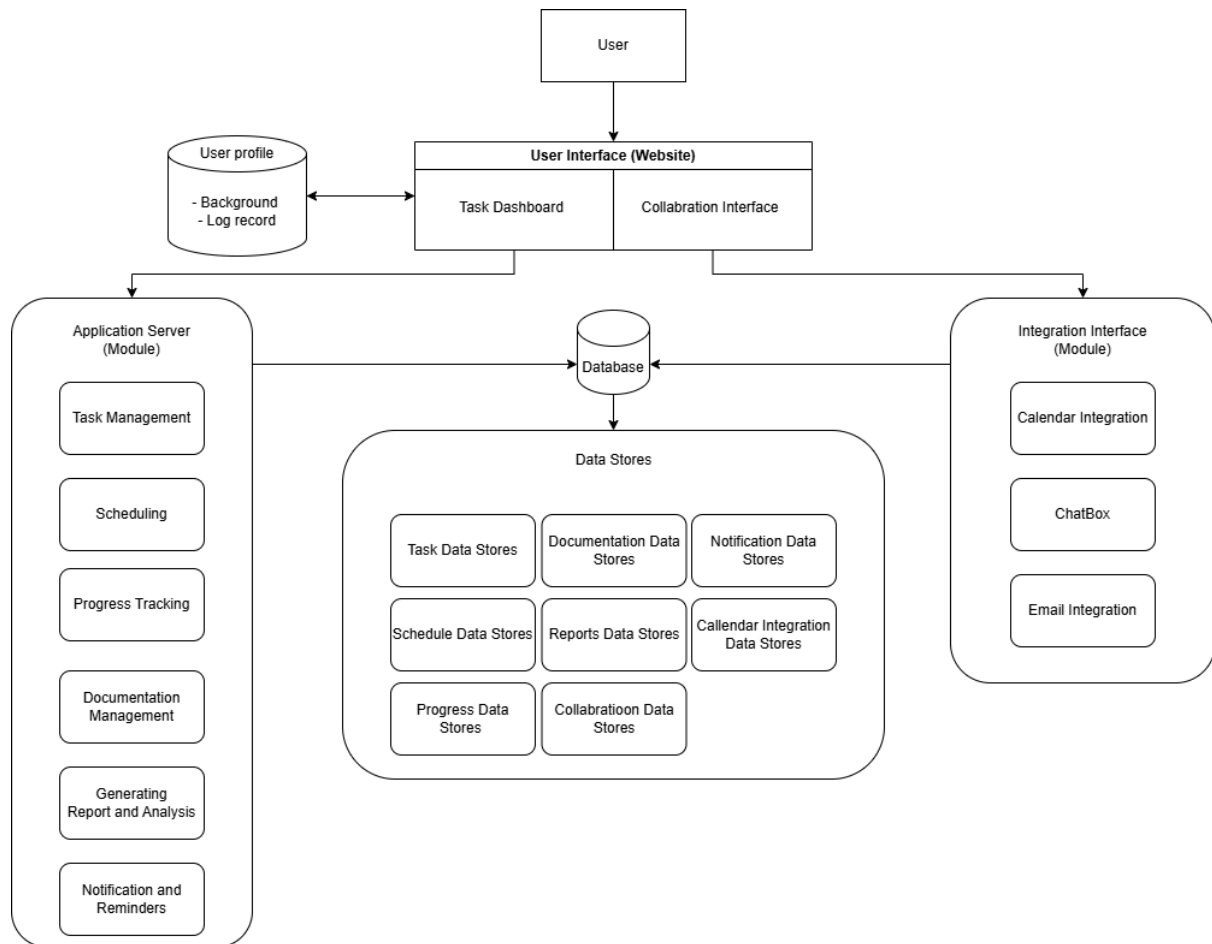
7.1.4 Event Response Table

Event	Source	Trigger	Activity	Response	Destination
User attempts to log in	User	User email and password	A user submits login credentials	Validate credentials and provide access or return error message	User
User updates profile information	User	User profile information	A user modifies personal information (e.g., name, email, language preference)	Update the user information in the database	User
User adds a new paper	User	Paper information	A user inputs details for a new paper	Create a new paper record in the database and save the paper details	User
User updates paper progress	User	Paper progress	A user logs progress updates for an existing paper	Update the progress information in the paper record and notify collaborators and supervisor	User
Deadline approaches for a paper	User	Upcoming deadline	The system detects an upcoming deadline for a paper	Send notification to the user and supervisor about the upcoming deadline	User
User sends a message to supervisor	User	Messages	A user initiates communication with their supervisor	Log the message in the chat log and notify the supervisor	Supervisor

Supervisor comments on paper progress	Supervisor	Supervisor's comments	The supervisor provides feedback on the paper progress	Update the paper record with supervisor comments and notify the	User
Paper is completed and approved by supervisor	Supervisor	Approval and report	The paper meets all requirements and gets the supervisor's approval	Mark the paper as complete, generate a report, and prepare for publication	User
User updates calendar entry	User	Modifies entry	A user adds or modifies an entry in the calendar	Update the calendar with new or modified deadlines and events	User
User generates a report	User	Report for paper statistics	A user requests a report generation for paper statistics	Retrieve relevant data from the database, compile the report, and present it to the user	User

This table outlines the key events that occur within the Task Management System and the corresponding responses by the system. For example, when a user attempts to log in, the system validates their credentials and grants access if the credentials are correct or returns an error message if they are not. When a user updates their profile or paper progress, the system updates the respective records and notifies relevant stakeholders as needed. Each event is matched with an appropriate system response to ensure the smooth functioning of the Task Management System.

7.1.5 System Architecture



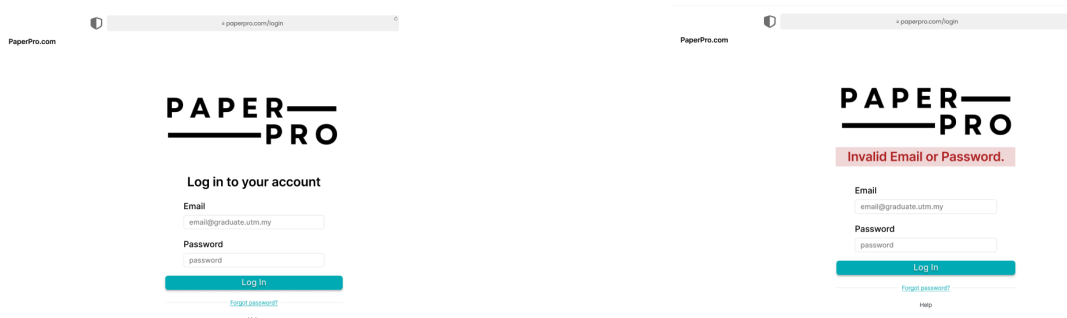
8.0 System Wireframe (Input Design, Output Design)

Figma prototype link:

<https://www.figma.com/proto/RmNT52oJomthnPl6MLhNyL/Untitled?t=wnjNmGhSlajBNL6H-1>

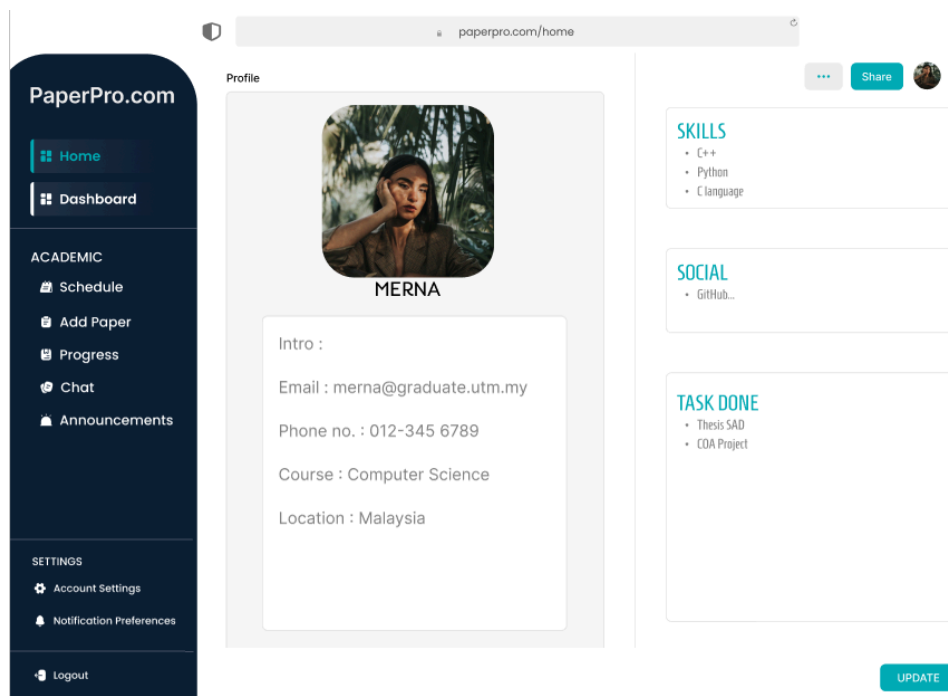
8.1 Process 1: Login

For the member account and password output, the users need to use their official school email address to log in to the system. For the login result output, if the user has logged in successfully, the website would automatically jump to the system's main page. Otherwise, the error message pops out, requesting the user to re-enter their login details.



8.2 Process 2: Manage Profile

Once the user logs in to the system, the user profile will be the main page. User's details will be shown in the profile for supervisor's understanding. Users can edit their profile if they find their information is wrong.



8.3 Process 3: Add and Manage Paper

The figure above is the example output that will show in process 2. In order to add paper, users need to enter paper details such as paper type, paper title, area of study and also collaborators information. The message will pop out when the paper is added successfully.

The figure displays a sequence of seven screenshots illustrating the 'Add Paper' process in the PaperPro.com application. The interface is divided into a sidebar with navigation options (Home, Dashboard, Academic, Schedule, Add Paper, Progress, Chat, Announcements) and a main content area for the 'Add Paper' form.

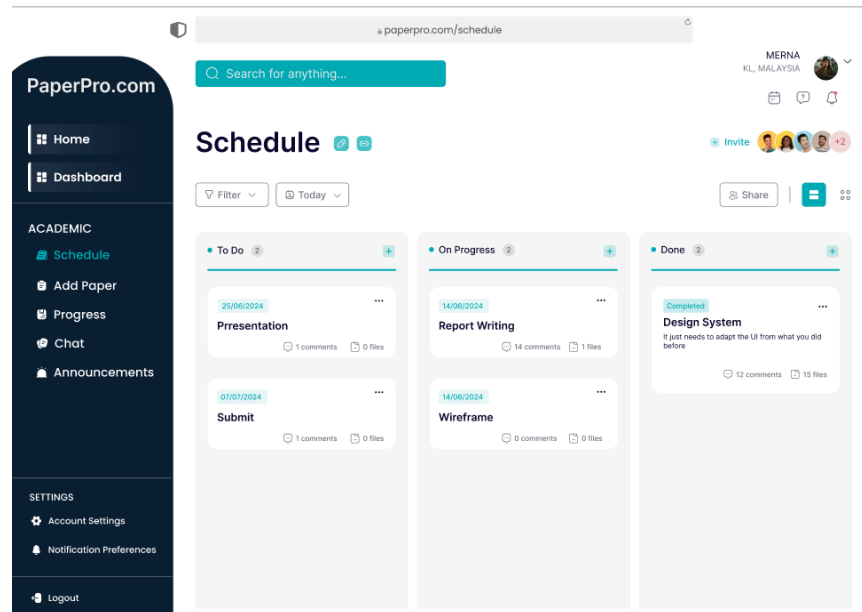
The form fields and their values across the screenshots are as follows:

- Screenshot 1:** Paper Title: [Empty], Area: [Empty], Paper Type: [Empty], Journal/Conferences: [Journal selected], ISI/Scopus: [ISI selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 2:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 3:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 4:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 5:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 6:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].
- Screenshot 7:** Paper Title: SAD Phase 3, Area: Computer Science, Paper Type: Research Paper, Journal/Conferences: [Journal selected], ISI/Scopus: [Scopus selected], Date: [YYYYMMDD], First Author: [Empty], Co Author: [Empty], Collaborator: [Empty].

The final screenshot shows a 'Notice' pop-up indicating 'Paper successfully saved'.

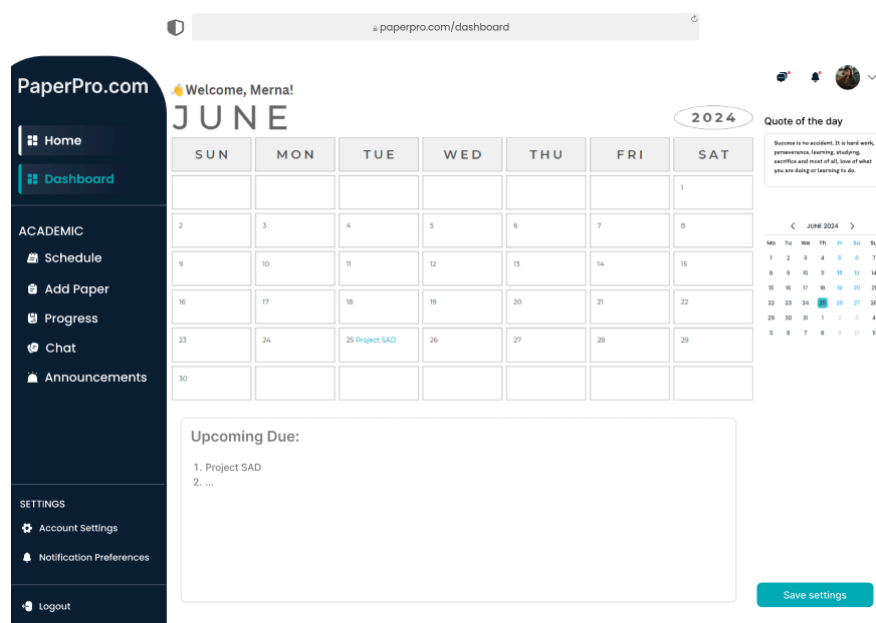
8.4 Process 4: Track and Update Progress

A Kanban board will be shown as in the figure above. Users need to always update their paper writing progress in the kanban board. At the same time, supervisors can track the user's writing progress at the kanban board.



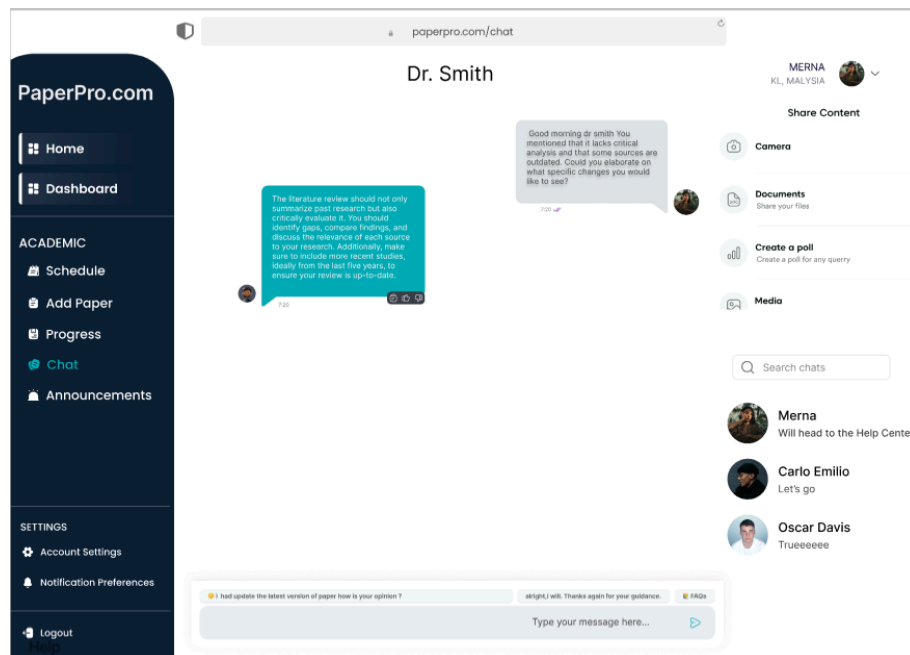
8.5 Process 5: Manage Calendar and Deadlines

A calendar will be displayed in the user's dashboard shown in the figure above. Users can add deadlines on the calendar. Users will get the deadline notifications for their paper writing on the dashboard.



8.6 Process 6: Communicate with Supervisor

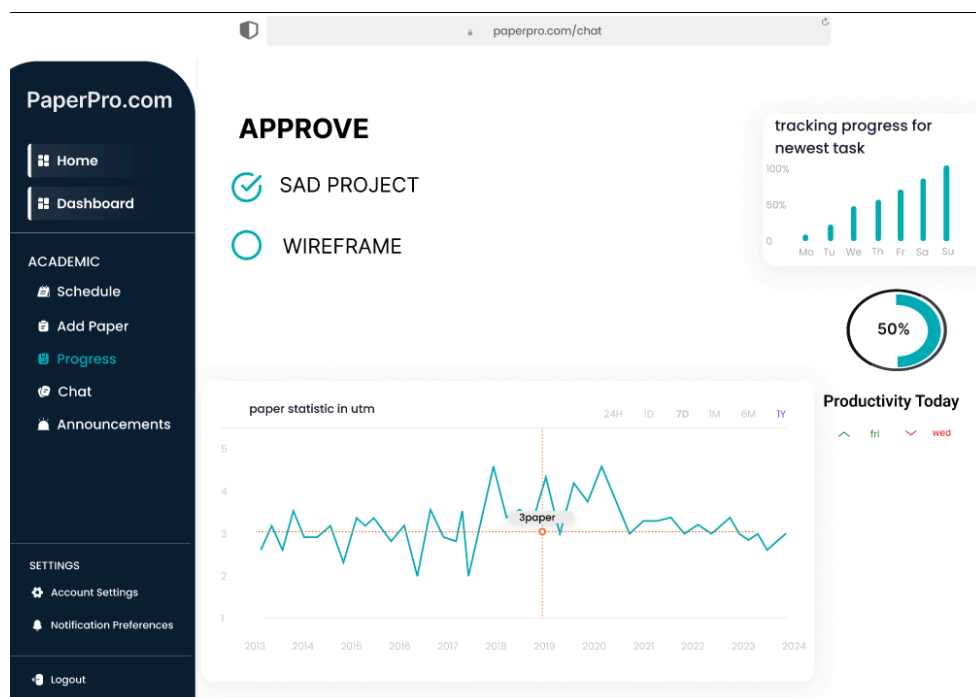
A chatbox will be provided for enabling the user to communicate with the supervisor if needed.



8.7 Process 7: Generate Reports

Users will submit the completed paper upon the deadlines and their papers need to be approved by their supervisor.

Once the paper gets approval from the supervisor, a report will be displayed.



9.0 Summary of proposed system

In conclusion, our group has completed the analysis and design for the Task Management System. In this phase which is project phase 3, we have done the Logical DFD TO-BE system, process specification, Physical DFD TO-BE system. In the AS-IS system, although it can be counted as a completed system, it may have some drawbacks, like users don't know the reason why supervisors reject their paper submission. The system also does not provide the supervisor with any information about the user (student) beforehand, leading to potential inefficiencies or misunderstandings in evaluating their work. The international students may face difficulties in using the system as they may not be familiar with the language. The TO-BE system has helped the user to solve the problem as stated above. The chat box is provided so that the user can communicate with the supervisor. The proposed system will present a brief student profile on the main page to provide supervisors with an overview of the student before they begin reviewing the student's paper. The system offers a wide range of languages to cater to the international students.